Path Re-planning

1. Course Content

- Basic

Run the sample program. Based on the road network planning and navigation, if an obstacle is detected on the traveling path during operation, re-planning will be triggered to find a new path in the road network.

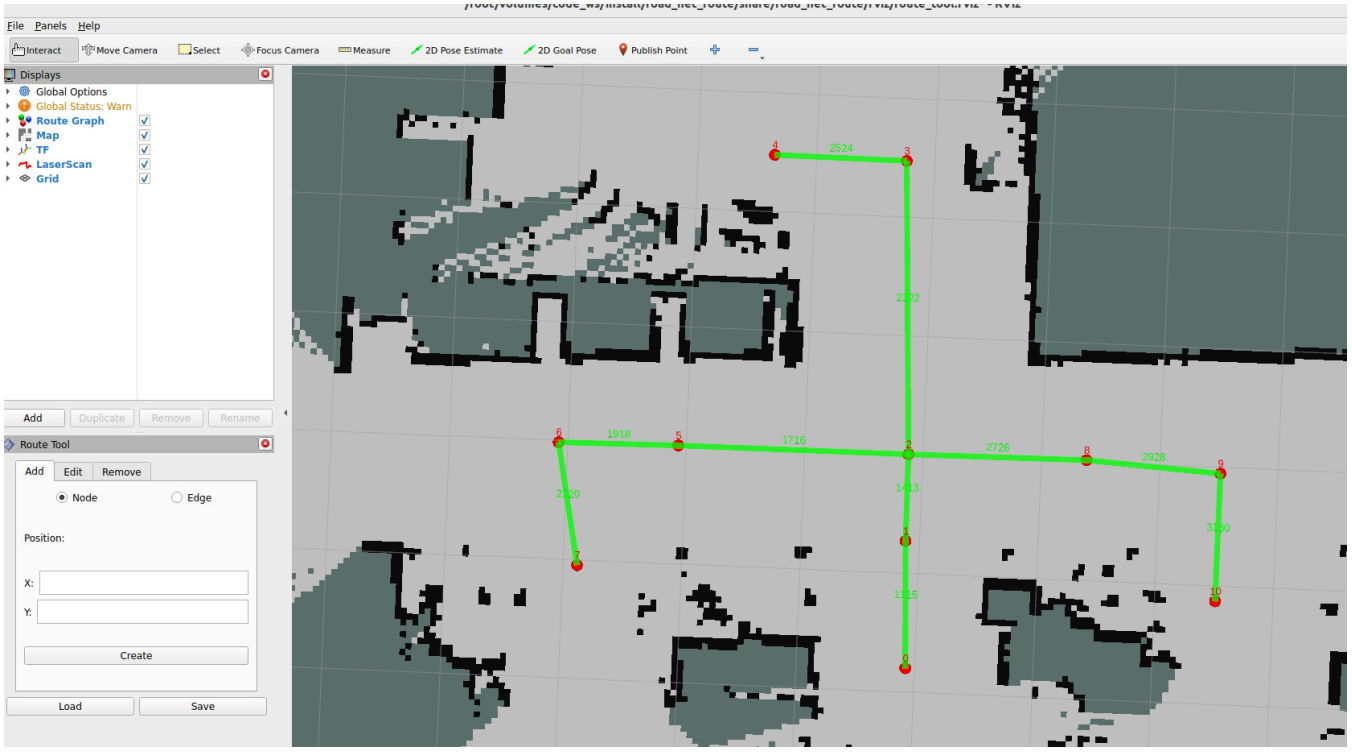
- Advanced

1. Understand the implementation method of re-planning and how to modify re-planning strategies (the strategy is not unique and can be implemented according to your needs).

2. Adjust the obstacle detection threshold based on the queried cost value.

2 Editing the Road Network

- To demonstrate the re-planning function, we add some edges to the sample road network from the previous course. For the operation tutorial on loading and editing the road network, refer to: **03-Road Network Annotation**: **Loading and Modifying Road Network Description** section.

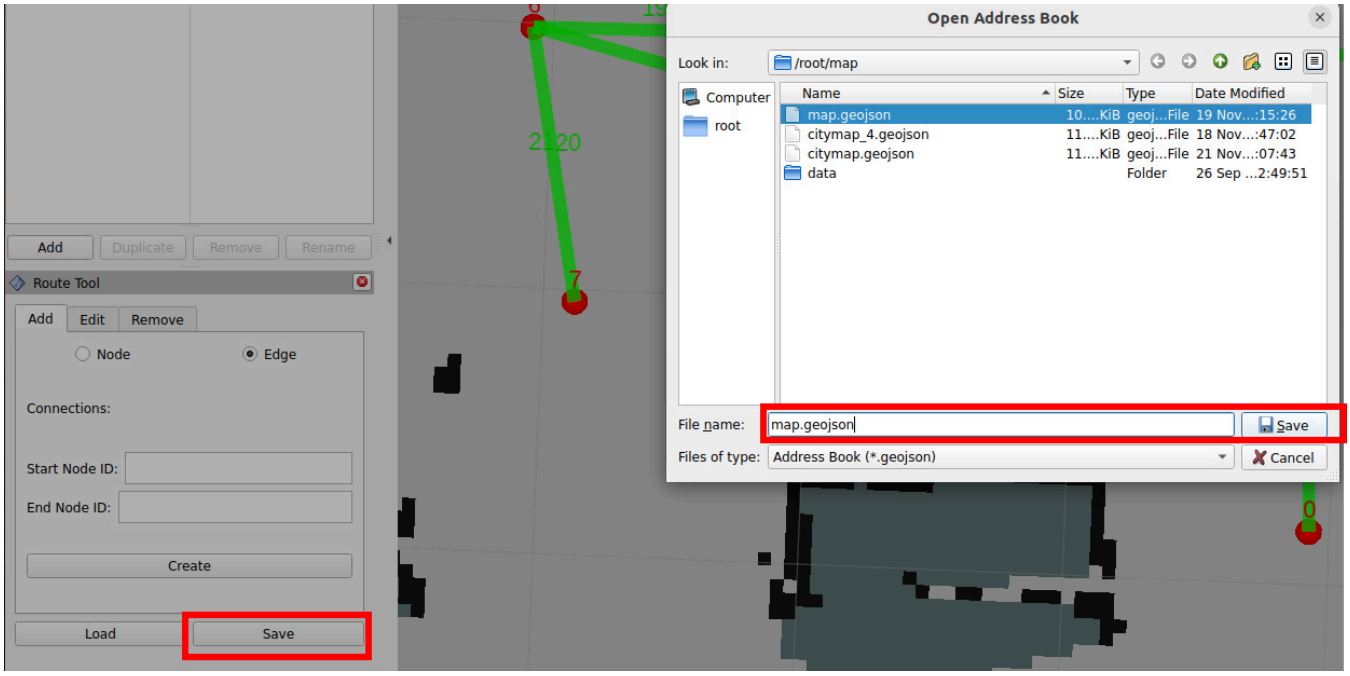


- For example, add a bidirectional edge between waypoint 1 and 6

- All operations will have corresponding records in the terminal

```
[rviz2-1] [INFO] [1764245498.392310926] [route_tool_node]: Adding edge from 1 to 6
[rviz2-1] [INFO] [1764245596.474192853] [route_tool_node]: Adding edge from 6 to 1
```

- Then save the file

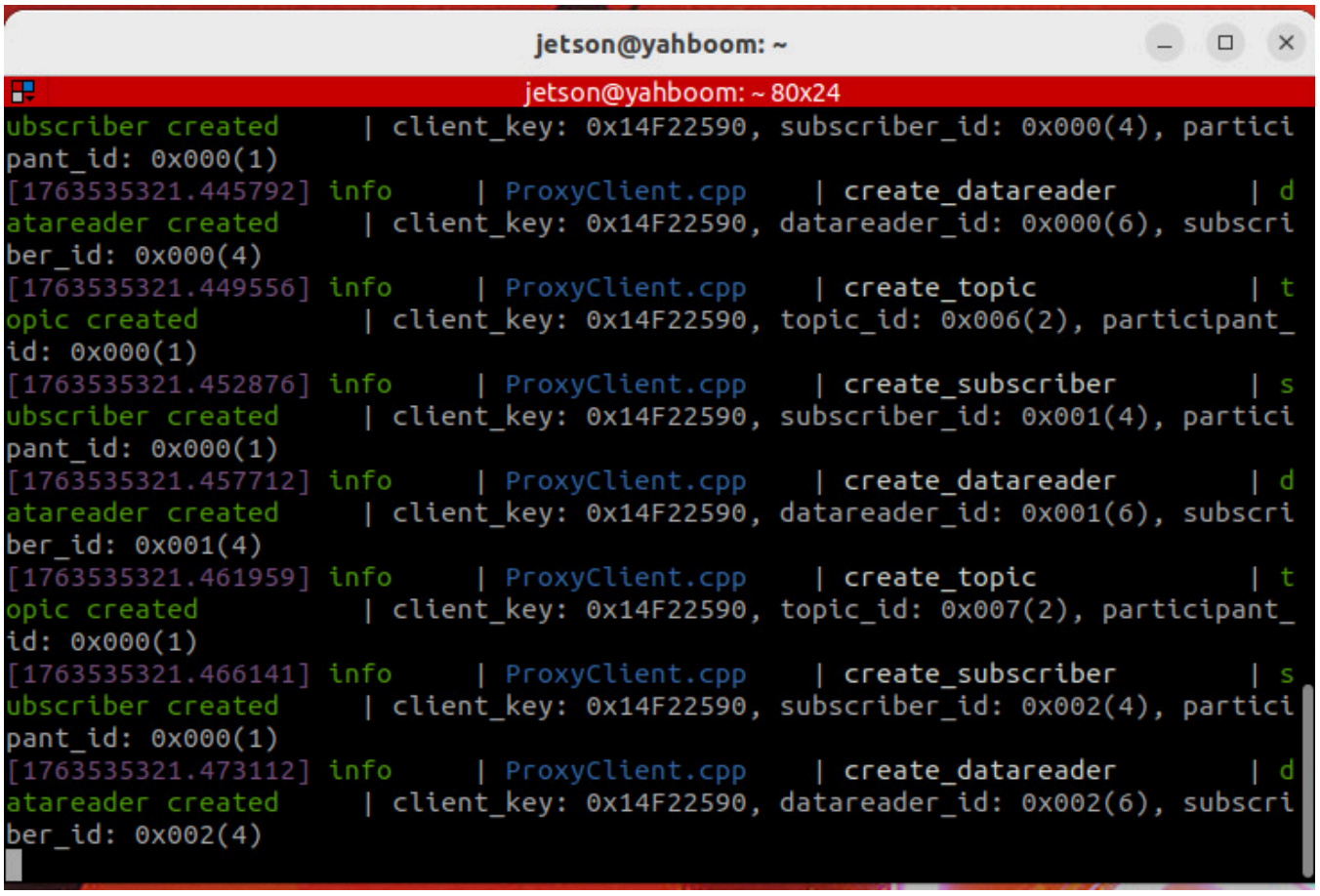


3. Running the Path Re-planning Case

3.1 Start the Agent

- Use the following command to start the microROS agent and connect to the chassis

```
sh start_agent.sh
```



- Start the road network container

```
roadnet start
```

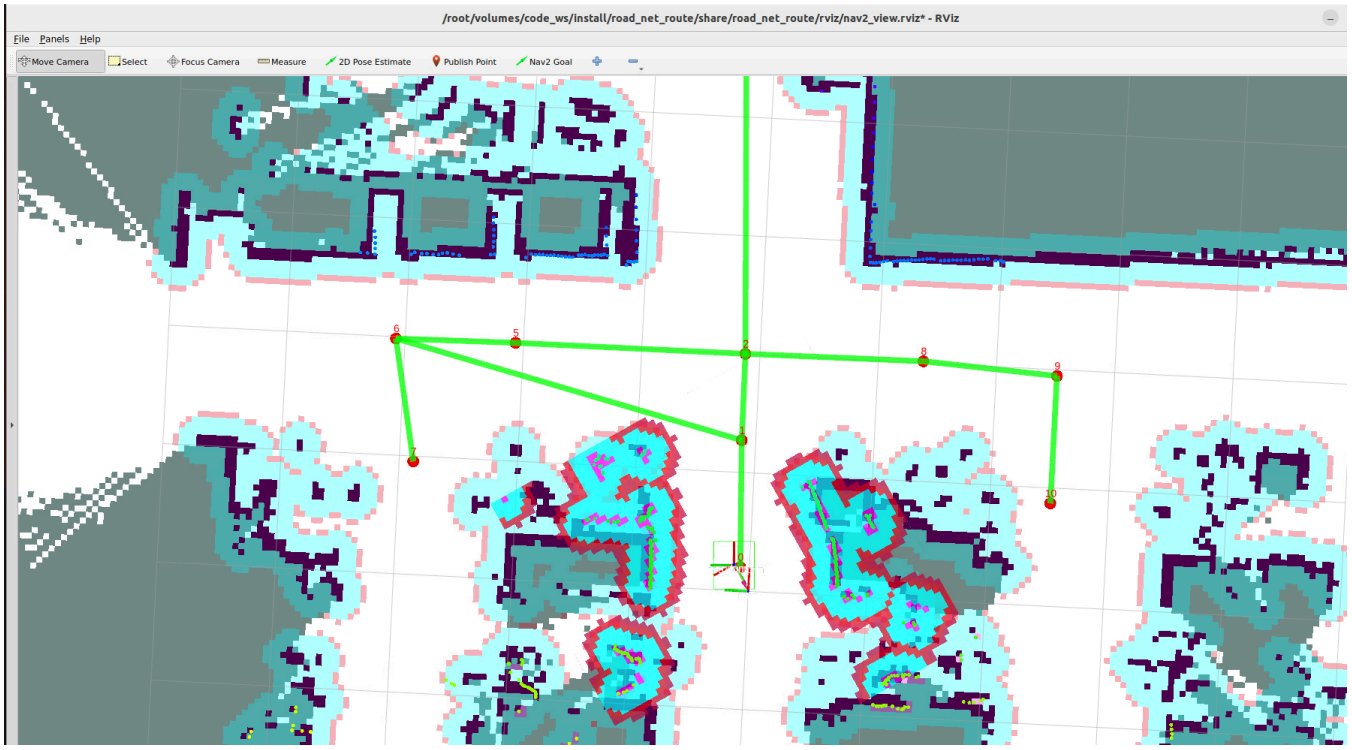
- Open a terminal on the host machine to start the LiDAR fusion node and odometry fusion (PI5 users should start in the `m3pro container`)

```
ros2 launch m3pro_bringup car_base.launch.py
```

- Start the road network route planning and navigation node (inside the roadnet container)

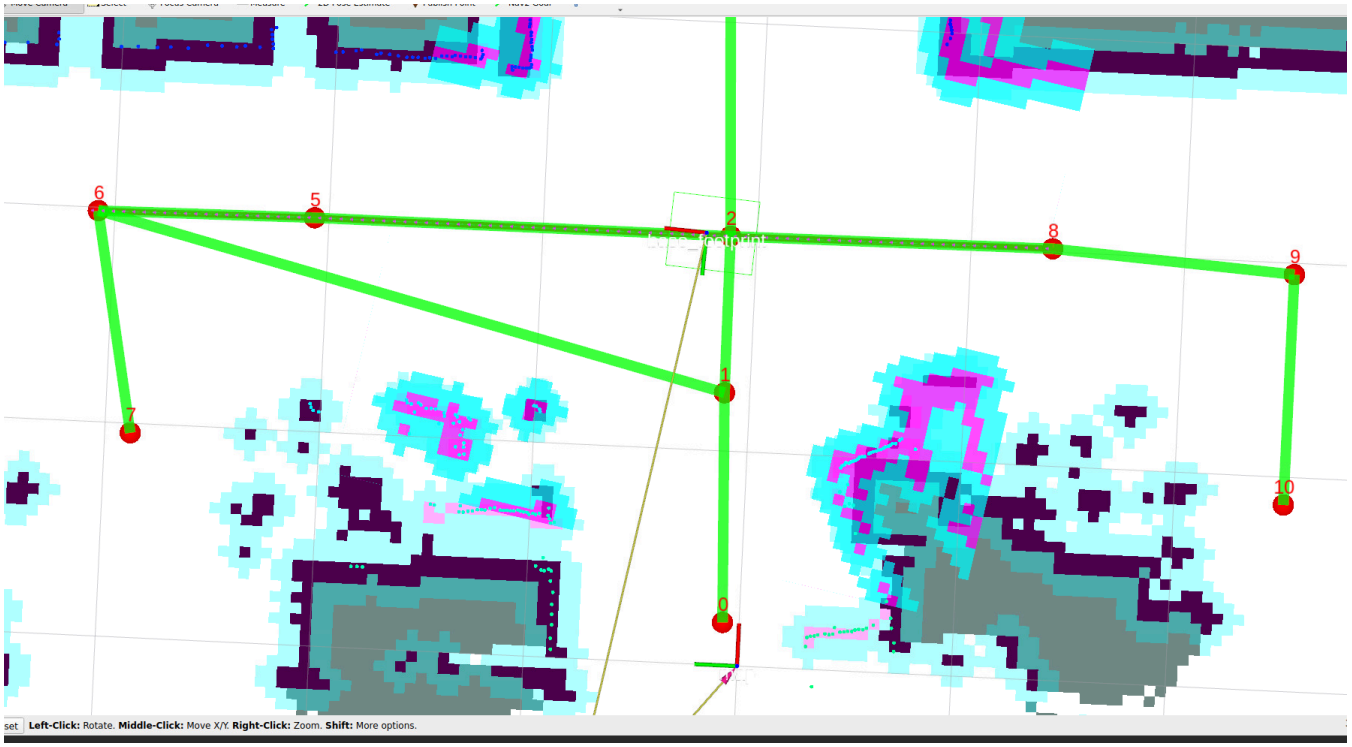
```
roadnet bash
```

```
ros2 launch road_net_route roadnet_nav2.launch.py
rviz_config_file:=/root/volumes/code_ws/src/road_net_route/rviz/nav2_relocaton.rviz
```



- The test scenario here is: travel from waypoint 8 to waypoint 6 along the road network, then place an obstacle on the planned path to trigger the re-planning strategy

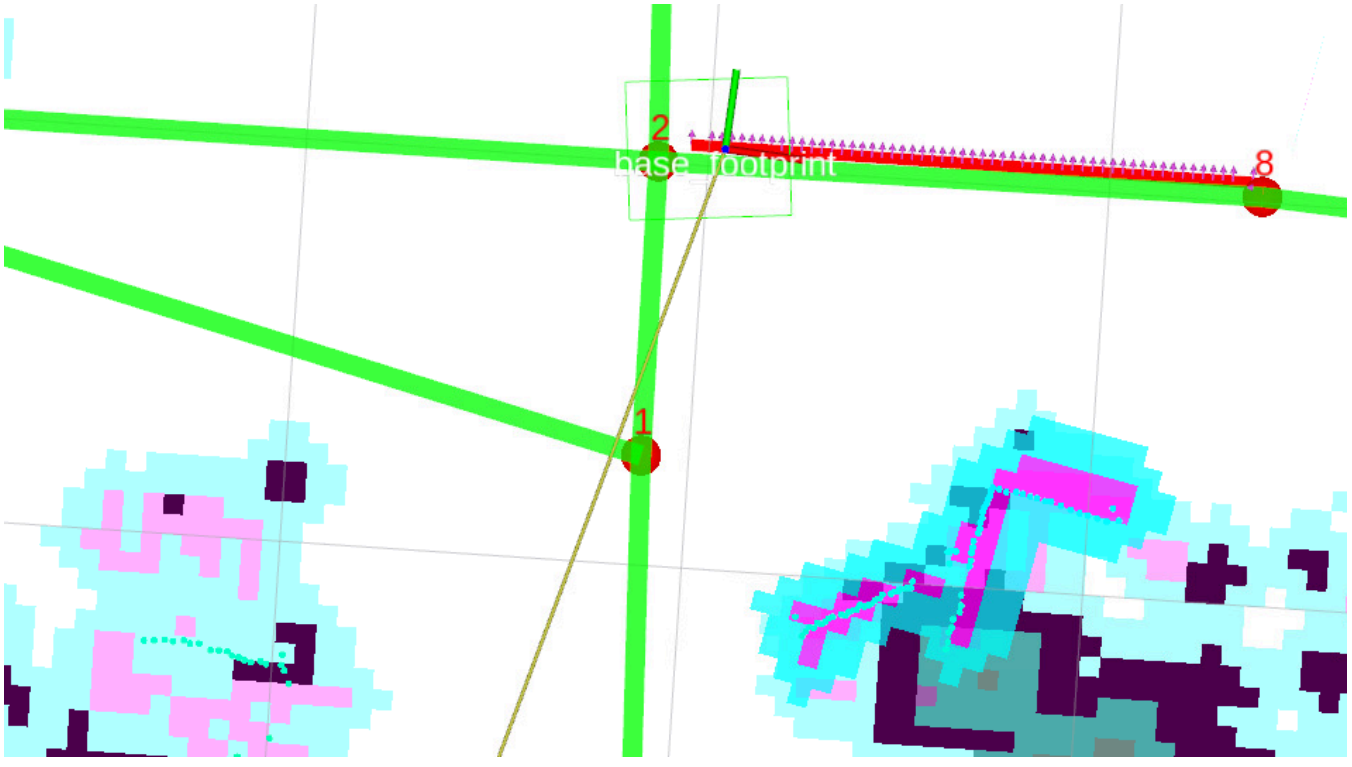
```
ros2 topic pub -1 /road_net_nav_id std_msgs/msg/Int16MultiArray "{data: [8, 6]}"
```



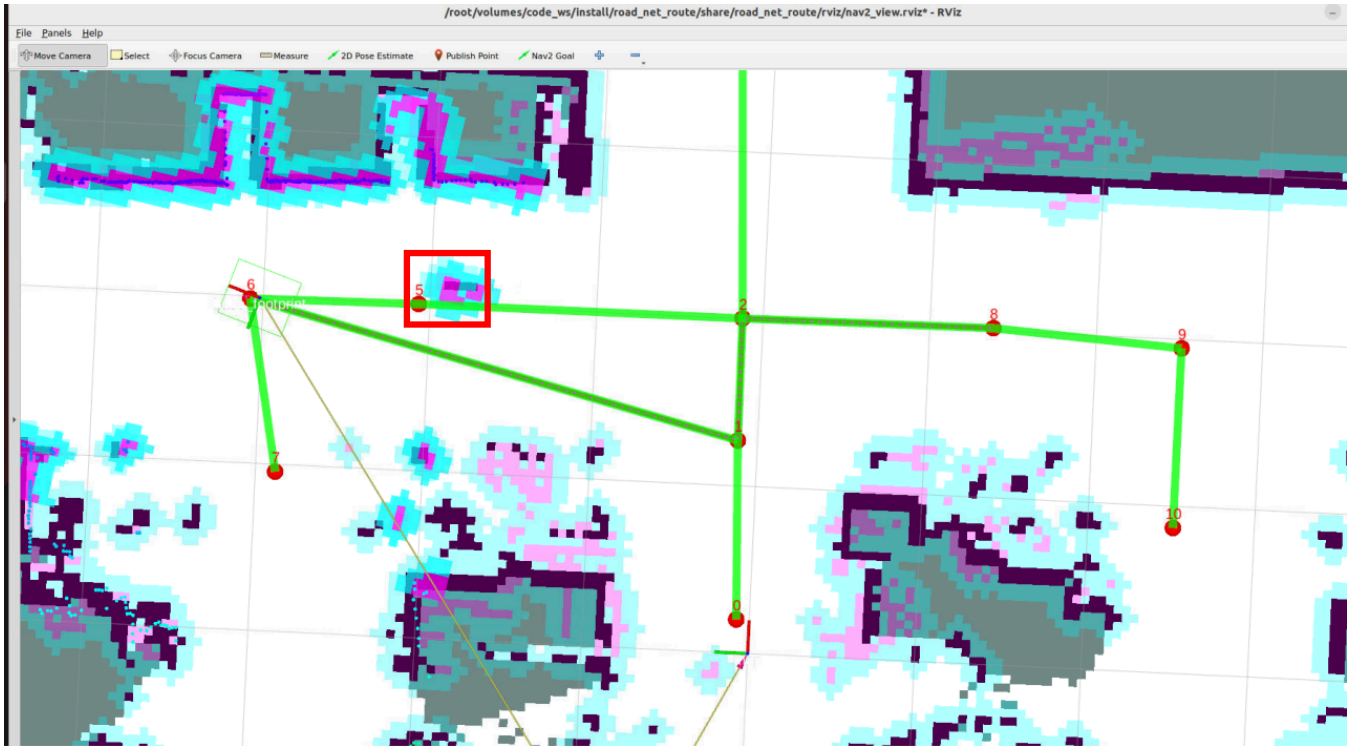
- Here you can see the terminal will prompt that there is an obstacle on the path, then trigger re-planning

```
[route_bridge-4] [INFO] [1764248055.001985898] [route_bridge]: id:2 ------> id:5 edge:16
[component_container_isolated-1] [INFO] [1764248055.687587193] [global_costmap.global_costmap]: StaticLayer: Resiz
ing costmap to 225 X 599 at 0.050000 m/pix
[route_bridge-4] [INFO] [1764248057.758462756] [route_bridge]: Path contains 7 potential obstacles.
[route_bridge-4] [INFO] [1764248058.249064867] [route_bridge]: Navigating to goal: 2.0027759075164795 0.1238360032
4392319...
```

- Here, after detecting an obstacle on the path, the path is impassable, and it retreats back to the previously passed waypoint 8



- Then from waypoint 8, try to plan a drivable path again and reach the target waypoint 6



4. Querying Obstacle Cost Threshold

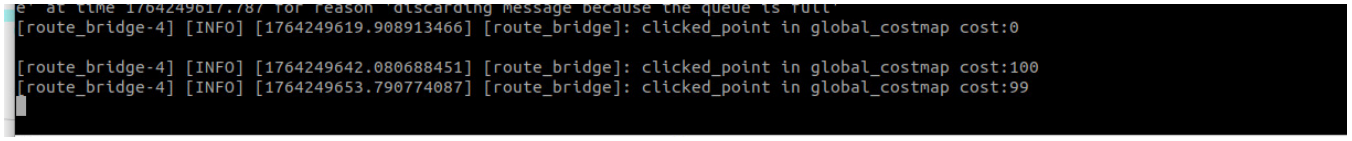
- The obstacle detection on the path is determined by detecting the cost value on the global path and the number of high-cost points on the global path. The threshold for triggering re-planning can be set by yourself.

> **Note**
>
> - The obstacle detection threshold is generally set to the factory default!!!
> - If there are special scenario requirements, you can refer to this part of the tutorial to adjust the threshold for triggering re-planning.
> - The cost value is the value used in the costmap to describe the area occupied by obstacles. The closer to the obstacle, the higher the cost value.

4.1 Viewing Obstacle Cost Values

- Click the Publish Point tool, then use the left mouse button to click on the location on the map where you want to view the cost value.

- You can see that areas without obstacles have a cost value of 0, and the cost value at the center of the obstacle is 99-100



4.2 Modifying the Threshold for Triggering Re-planning

- Configuration file path

```
$HOME/M3Pro_ws/RoadNetwork/volumes/code_ws/src/road_net_route/config/nav2_roadnet.yaml
```

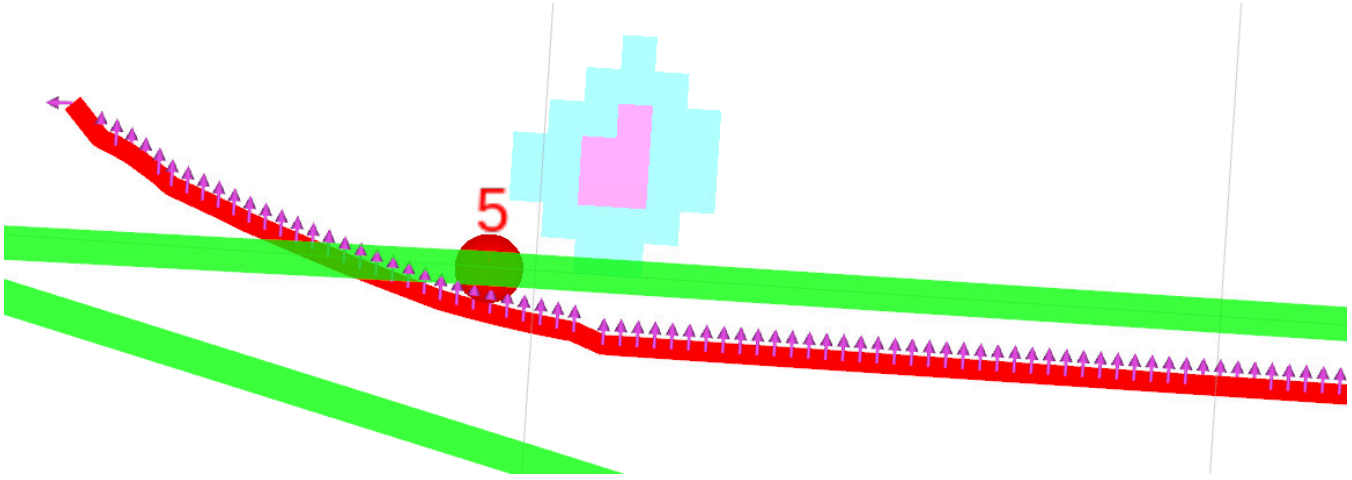
- Find the route_bridge node configuration parameters in the configuration file, where `COST_THRESHOLD` and `PATH_OBSTACLES_THRESHOLD` together determine the triggering threshold.

```yaml
route_bridge:
  ros__parameters:
    roadnet_file: '/root/map/map.geojson'
    COST_THRESHOLD: 60
    replace2id: True
    PATH_OBSTACLES_THRESHOLD: 5
```

- The specific obstacle detection logic is as follows:
- The global path planned by the road network function is the `nav_msgs/msg/Path` message type in ROS, which consists of multiple pose points. When performing road network navigation, all pose points on the global path are checked whether they have high-cost values in the global costmap.
    - If the cost value of a pose point on the path exceeds the **COST_THRESHOLD**, that point on the

path is marked as a high-cost pose point.
    - Finally, calculate how many pose points on the global path are high-cost pose points.
    - If the total number of high-cost pose points is greater than **PATH_OBSTACLES_THRESHOLD**, the path is considered impassable and re-planning needs to be performed.

> **Tip**
>
> If obstacle detection is not sensitive enough, you can increase **COST_THRESHOLD** and **PATH_OBSTACLES_THRESHOLD**. If it is too sensitive, you can decrease these two values or adjust the obstacle inflation radius (for modifying the obstacle inflation radius, refer to 04-Road Network Route Planning and Navigation section 4.1 Navigation Controller Debugging).



5. Source Code Analysis

- Implementation source code path:

```
$HOME/M3Pro_ws/RoadNetwork/volumes/code_ws/src/road_net_route/road_net_route/route_bridge.py
```

5.1 Obstacle Detection and Re-planning Strategy

- Iterate through each pose point of the global path, check whether there are obstacles exceeding the threshold on the path through the global costmap, and return a boolean value indicating whether the path is blocked.
- **World coordinates to map coordinates**: Call the costmap's `worldToMapValidated` method to convert world coordinates `(wx, wy)` to grid coordinates `(mx, my)` of the costmap (the costmap is a 2D grid map, each grid corresponds to a cost value).
- **Coordinate validity check**: If the converted grid coordinates are `None`, it means that path point is outside the global costmap range, a warning log is printed, and that point is skipped.
- **Parameters**: `path:Path` → ROS2's `nav_msgs/msg/Path` message type, storing the robot's global

planned path, containing a series of poses (`poses`).

- **Return value**: `bool` → `True` means the number of obstacles on the path exceeds the threshold, `False` means the path is feasible.

```
def __check_path_obstacles(self, path:Path)->bool:
    """Check for obstacles on the global path."""
    pose_array=[]
    for pose in path.poses:
        wx=pose.pose.position.x
        wy=pose.pose.position.y

        mx, my = self.global_costmap.worldToMapValidated(wx,wy )
        if mx is None or my is None:
            self.get_logger().warn(f"Path point ({wx}, {wy}) is outside global costmap")
            continue

        cost = self.global_costmap.getCostXY(int(mx),int(my))

        if cost > self.COST_THRESHOLD:
            pose_array.append(pose)
    if len(pose_array) >self.PATH_OBSTACLES_THRESHOLD:
        self.get_logger().info(f"Path contains {len(pose_array)} potential obstacles.")
        return True
    return False
```

- After detecting obstacles on the global path, let the robot **retreat to the previous waypoint** and re-plan the path.

```
def __reroute_strategy_1(self,last_id):
    '''Replanning Strategy 1: Go back to the previous waypoint and replan the drivable
path.'''
    navigator= BasicNavigator()
    goal_x,goal_y=self.__get_node_coordinates(last_id)
    if goal_x is not None and goal_y is not None:
        goal_pose = PoseStamped()
        goal_pose.header.frame_id = "map"
        goal_pose.pose.position.x = goal_x
        goal_pose.pose.position.y = goal_y
        back_lastid = navigator.goToPose(goal_pose)
        while not navigator.isTaskComplete(task=back_lastid):
            time.sleep(0.1)
        if navigator.getResult() == TaskResult.SUCCEEDED:
            self.get_logger().info("back to last id  success.")
        else:
            self.get_logger().info("back to last id  failed.")
```

- handle_id_navigation and handle_pose_navigation are the core code responsible for road network

planning and navigation. The following parts in these two functions contain the logic for detecting paths and triggering re-planning.

When the path is detected to be infeasible, call **Re-planning Strategy 1**: let the robot **retreat to the previous waypoint** (the coordinates corresponding to `last_node_id`). The core logic of this method has been explained earlier: initialize the navigator → get the coordinates of `last_node_id` → send a navigation task of "retreat to this waypoint" → wait for the task to complete and print the result. Withdraw the robot from the current obstacle area to a **safe predefined waypoint**, providing a stable starting point for subsequent path re-planning.

The complete flow is as follows:

1. The robot normally executes path tracking, obtaining navigation feedback in real-time.
2. Check whether there are obstacles in the path from `feedback`:
    - If no obstacles: continue executing the original path, no processing needed.
    - If there are obstacles: trigger the re-planning logic.
3. The robot retreats to the previous safe waypoint (`last_node_id`).
4. Using this waypoint as the starting point, re-plan the path to the target waypoint (`goal_id`).
5. Obtain feedback from the new path and let the robot continue moving while tracking the new path.

```
if feedback is not None and self.__check_path_obstacles(feedback.path):
    self.__reroute_strategy_1(last_node_id)
    route_task = self.navigator.getAndTrackRoute(last_node_id, goal_id)
    feedback = self.navigator.getFeedback(task=route_task)
    follow_path_task = self.navigator.followPath(feedback.path)
```

```
    def handle_id_navigation(self, start_id:int, goal_id:int):
        '''Road network ID navigation'''

        route_task = self.navigator.getAndTrackRoute(start_id, goal_id)
        feedback=None
        follow_path_task=RunningTask.NONE
        last_feedback = None
        last_node_id =999
        while not self.navigator.isTaskComplete(task=route_task):
            feedback = self.navigator.getFeedback(task=route_task)
            if feedback is not None and self.__check_path_obstacles(feedback.path):
                self.__reroute_strategy_1(last_node_id)
                route_task = self.navigator.getAndTrackRoute(last_node_id, goal_id)
                feedback = self.navigator.getFeedback(task=route_task)
                follow_path_task = self.navigator.followPath(feedback.path)
            while feedback is not None:
                if  last_feedback is not None and (feedback.last_node_id !=
last_feedback.last_node_id or feedback.next_node_id != last_feedback.next_node_id):
                    self.get_logger().info(f'id:{feedback.last_node_id} ------> id:
{feedback.next_node_id} edge:{feedback.current_edge_id}')
                last_feedback = feedback
```

```
                last_node_id=feedback.last_node_id
                if feedback.rerouted :
                    self.get_logger().info('reroute a new path to follow!')
                    follow_path_task = self.navigator.followPath(feedback.path)
                feedback = self.navigator.getFeedback(task=route_task)

            if self.navigator.isTaskComplete(task=follow_path_task):
                print(follow_path_task)
                self.get_logger().info('Controller completed its task!')
                if follow_path_task != RunningTask.NONE :
                    self.navigator.cancelTask()

        result = self.navigator.getResult()
        if result == TaskResult.SUCCEEDED:
            self.get_logger().info(Fore.GREEN+"navigation task completed."+Fore.RESET)
            self.action_feedback_pub.publish(String(data="road_net_nav_succeeded"))
        elif result == TaskResult.CANCELED:
            self.get_logger().warn("Navigation task canceled.")
        elif result == TaskResult.FAILED:
            self.get_logger().error("Navigation task failed.")
            self.action_feedback_pub.publish(String(data="road_net_nav_failed"))

    def handle_pose_navigation(self, goal_pose:PoseStamped):
        '''Arbitrary pose navigation'''
        transform = self.tf_buffer.lookup_transform("map", "base_footprint",
rclpy.time.Time(),rclpy.duration.Duration(seconds=5.0))
        if transform is None:
            self.get_logger().info(Fore.RED+"Failed to get TF transform "+Fore.RESET)
            return

        else:
            start_pose = PoseStamped()
            start_pose.header = transform.header
            start_pose.pose.position.x = transform.transform.translation.x
            start_pose.pose.position.y = transform.transform.translation.y
            start_pose.pose.orientation = transform.transform.rotation
            self.get_logger().info(Fore.CYAN + f"current:
{transform.transform.translation}"+Fore.RESET)
            nearest_start_node_id, nearest_start_distance =
self.__find_nearest_node(start_pose.pose.position.x, start_pose.pose.position.y)
            nearest_goal_node_id, nearest_goal_distance =
self.__find_nearest_node(goal_pose.pose.position.x, goal_pose.pose.position.y)
            if self.replace2id:
                self.get_logger().info(Fore.CYAN + f"nearest_start_node_id:
{nearest_start_node_id}, distance:{nearest_start_distance}"
                                       f"\nnearest_goal_node_id:{nearest_goal_node_id},
distance:{nearest_goal_distance}"+Fore.RESET)
                route_task =
self.navigator.getAndTrackRoute(start=nearest_start_node_id,goal=nearest_goal_node_id)
            else:
                route_task =
self.navigator.getAndTrackRoute(start=start_pose,goal=goal_pose,use_start=True)
```

```
            feedback=None
            follow_path_task=RunningTask.NONE
            last_feedback = None
            last_node_id =999
            while not self.navigator.isTaskComplete(task=route_task):
                feedback = self.navigator.getFeedback(task=route_task)
                if feedback is not None and self.__check_path_obstacles(feedback.path):
                    self.__reroute_strategy_1(last_node_id)
                    route_task = self.navigator.getAndTrackRoute(last_node_id,
nearest_goal_node_id)
                    feedback = self.navigator.getFeedback(task=route_task)
                    follow_path_task = self.navigator.followPath(feedback.path)
                while feedback is not None:
                    if  last_feedback is not None and (feedback.last_node_id !=
last_feedback.last_node_id or feedback.next_node_id != last_feedback.next_node_id):
                        self.get_logger().info(f'id:{feedback.last_node_id} ------> id:
{feedback.next_node_id} edge:{feedback.current_edge_id}')
                    last_feedback = feedback
                    last_node_id=feedback.last_node_id
                    if feedback.rerouted :
                        self.get_logger().info('reroute a new path to follow!')
                        follow_path_task = self.navigator.followPath(feedback.path)
                    feedback = self.navigator.getFeedback(task=route_task)

                if self.navigator.isTaskComplete(task=follow_path_task):
                    self.get_logger().info('Controller completed its task!')
                    if follow_path_task != RunningTask.NONE :
                        self.navigator.cancelTask()

            result = self.navigator.getResult()
            if result == TaskResult.SUCCEEDED:
                if self.__adjust_pose(goal_pose):
                    self.get_logger().info(Fore.GREEN+"navigation task
completed."+Fore.RESET)
                    self.action_feedback_pub.publish(String(data="road_net_nav_succeeded"))

            elif result == TaskResult.CANCELED:
                self.get_logger().info(Fore.YELLOW+"navigation task canceled."+Fore.RESET)

            elif result == TaskResult.FAILED:
                self.get_logger().info(Fore.RED+"navigation task failed."+Fore.RESET)
                self.action_feedback_pub.publish(String(data="road_net_nav_failed"))
```

5.2 Query Map Cost Value Implementation Method

- Call the global costmap's `worldToMapValidated` method to convert **world coordinates `(x, y)` to grid coordinates `(mx, my)` of the costmap**.
- The costmap is a **2D grid map** (each grid corresponds to a small area in the real world, 0.05m×0.05m). `worldToMapValidated` is a safe conversion method: if the coordinates are outside the costmap range, it returns `None`, avoiding boundary errors caused by direct conversion.

- **Obtain cost value**: Call the costmap's `getCostXY` method, passing the integer form of the grid coordinates (grid coordinates are discrete integers, avoiding errors caused by floating-point input), to obtain the **cost value** `cost` of that grid.

```
    def get_costmap_value(self, msg:PointStamped):
        '''In RViz, you can click to view the cost of a point on the global cost map.'''
        x=msg.point.x
        y=msg.point.y
        mx, my = self.global_costmap.worldToMapValidated(x, y)
        if mx is None or my is None:
            self.get_logger().warn(f"Path point ({x}, {y}) is outside local costmap")
            return
        cost = self.global_costmap.getCostXY(int(mx),int(my))
        self.get_logger().info(f"clicked_point in global_costmap cost:{cost}")
```